

PROBLEM STATEMENT DETAILS

Problem Statement ID	26126
Problem Statement Title	Vision Based Autonomous Navigation for Unmanned Ground Vehicle for Outdoor environment
Description	Background Outdoor Unmanned Ground Vehicles (UGVs) face unpredictable terrain, changing light, and unreliable GPS signals. To achieve true autonomy in applications like search-and-rescue, agriculture, or delivery, UGVs must rely on onboard computer vision. Visual perception provides a cost-effective, data-rich way for vehicles to understand and safely navigate complex, unstructured outdoor surroundings. Description The objective is to build an autonomous navigation system for a UGV operating in a GPS-denied outdoor environment using camera feeds as the primary sensor. Students must solve three key challenges: • Path Detection: Real-time identification of safe, traversable paths vs. hazards (e.g., rocks, ditches, trees). • Visual Localization: Estimating the UGV's position and orientation without GPS using visual data. • Collision Avoidance: Dynamically routing the vehicle around sudden obstacles toward a destination. Expected Solution A functional software module consisting of: • Perception AI: A lightweight model for obstacle and path detection. • Visual SLAM/Odometry: A pipeline to track vehicle movement. • Path Planner: An algorithm to translate visual data into wheel/motor commands. Success Criteria: Successful, collision-free navigation from Point A to Point B across outdoor scenarios.
Organization	Bharat Electronics Limited
Department	Bharat Electronics Limited
Category	Software
Theme	Smart Automation
Youtube Link	
Dataset Link	
Contact info	